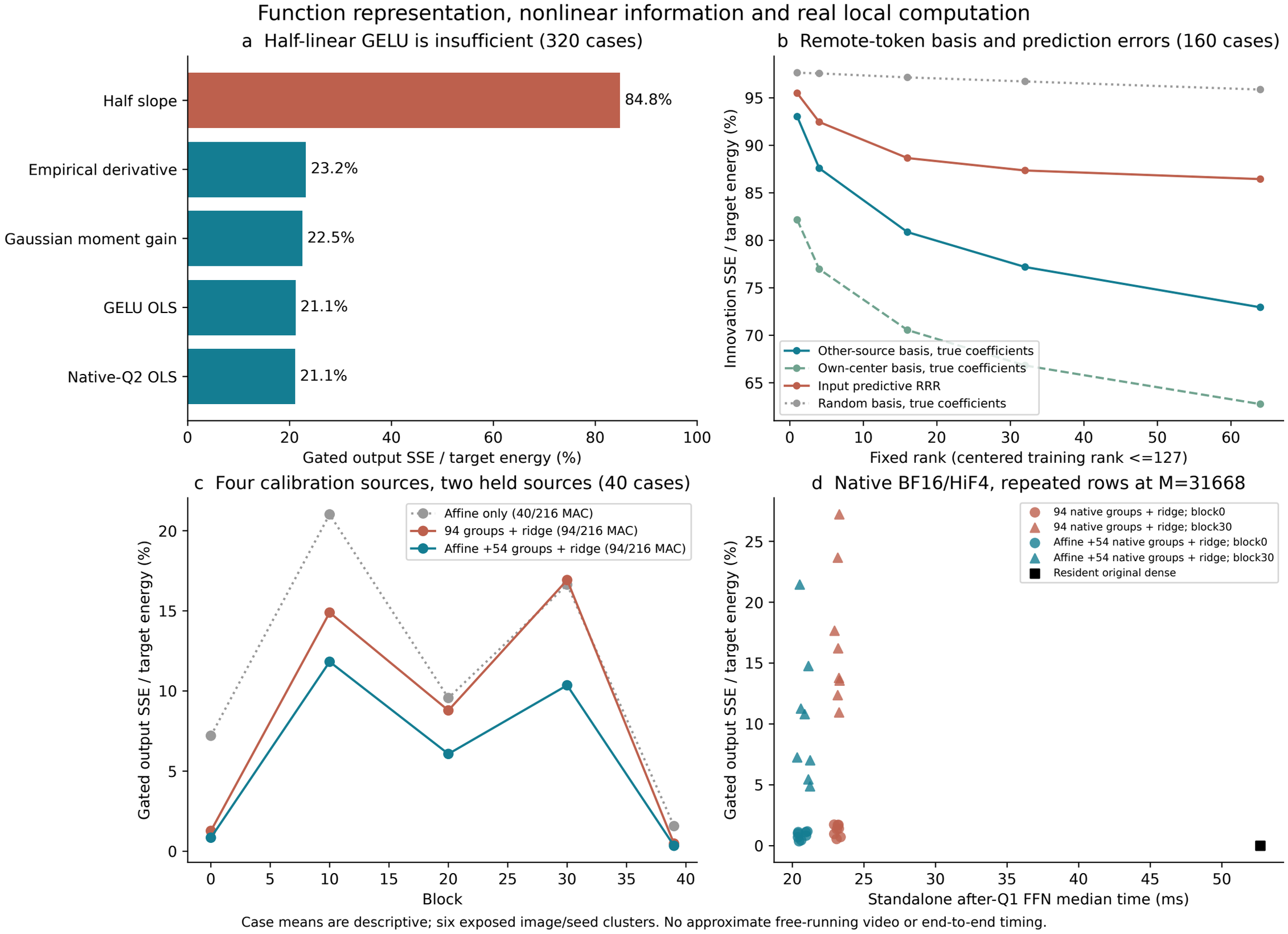
Function representation, nonlinear information and real local computation
a Half-linear GELU is insufficient (320 cases)
Half slope
Empirical derivative
Gaussian moment gain
GELU OLS
Native-Q2 OLS
84.8%
23.2%
22.5%
21.1%
21.1%
Gated output SSE / target energy (%)
b Remote-token basis and prediction errors (160 cases)
Innovation SSE / target energy (%)
Other-source basis, true coefficients
Own-center basis, true coefficients
Input predictive RRR
Random basis, true coefficients
Fixed rank (centered training rank <=127)
c Four calibration sources, two held sources (40 cases)
Affine only (40/216 MAC)
94 groups + ridge (94/216 MAC)
Affine +54 groups + ridge (94/216 MAC)
Gated output SSE / target energy (%)
Block
d Native BF16/HiF4, repeated rows at M=31668
94 native groups + ridge; block0
94 native groups + ridge; block30
Affine +54 native groups + ridge; block0
Affine +54 native groups + ridge; block30
Resident original dense
Gated output SSE / target energy (%)
Standalone after-Q1 FFN median time (ms)
Case means are descriptive; six exposed image/seed clusters. No approximate free-running video or end-to-end timing.